Artificial Intelligence. CSA1709

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1. Intelligent Agent in smart Home Automation.

Introduction:

An Intelligent agent is a system that observes its Environment through sensors and Perform action using actuators to achieve a goal.

Simple Reflex Agent :-

Acts based on current input.

Eg: switches on lights when motion is detected

It is fast but cannot remember past events.

Advantage: Simple, Fast, and Easy to implement.

Disadvantage: Cannot remember past event (or) Learn user's preferences.

Model - Based Agent:

Uses current input and past information

Eg: Switches off lights when nobody is in the room.

Goal - Based Agent :

A goal - based agent selects action to achieve a specific goal.

Eg: It adjust the air condition to maintain the desired room temperature.

Advantage :- Support planning & Goal - oriented decision

A utility based agent selects the action that Provides the greatest overall benefit by considering different factors.

Eg: It maintain a Comfortable temperature while reducing electricity consumption & maintaing security.

Advantage : Balances Comfort, energy efficiency & security. designing a suitable utility function is difficult.

2) Uniform Cost Search (US) & Iterative Deepening Search (IDS) in a Maze.

Introduction:-

Uninformed search algorithms explore possible Path without using additional information about the distance to the Goal.

Uniform Cost Search (UCS).

UCS expands the node with the lowest total path cost.

It uses a priority queue to select the next node.

Working:

Start from the initial position.

Select the Lowest - cost path.

Check for the Exit

Repeat Until the Exit is found.

Advantage:- Finds the least-Cost Path

Disadvantage:- Requires more memory and time

Iterative Deepening Search (IDS)

IDS repeatedly searches maze by increasing the depth limit until the exit is found.

Advantage:

Uses less memory and finds the shortest solution when step cost are Equal.

Disadvantage :- Repeats searches & may waste time

Complexity :-

UCS :- Exponential time & space in the general case :-

IDS :- Time : $(O(b^d)$, Space $(O(b))$

Here (b) is the branching factor & (d) is the solution depth.

Agent:

Simple :- Acts on current input

Model - Based :- Remembers visited positions.